Exercise JA

Q1. The function $f : [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

(A) one-one and onto (B) onto but not one - one (C) neither one- one nor onto

Ans. -

Sol. For real x, $f(x) = x^3 + 5x + 1$

$\lim\limits_{x \to \infty} f(x) = + \infty$

and $\lim\limits_{x \to \infty} f(x) = - \infty$

$\therefore$ Range is R – f (x) is on to

Now $f'(x) = 3x^2 + 5 > 0$

$\therefore$ f(x) is one-one

f(x) is one-one onto

Q2. Let $f: (-1, 1) \rightarrow R$ be such that $f(\cos 4\theta) = \dfrac{2}{2 - \sec^2 \theta}$ for $\theta \in \left(0, \dfrac{\pi}{4}\right) \cup \left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$. Then the value(s) of $f\left(\dfrac{1}{3}\right)$ is (are)

(A) $1 - \sqrt{\dfrac{3}{2}}$ (B) $1 + \sqrt{\dfrac{3}{2}}$ (C) $1 - \sqrt{\dfrac{2}{3}}$ (D) $1 + \sqrt{\dfrac{2}{3}}$

Ans. -

Sol. $f(x) = (x + 1)^2 - 1; \geq x - 1$

$f'(x) = 2(x + 1) \geq 0$ for $x \geq -1$

f(x) is bijection

Statement (2) is correct

$\therefore$ Now $f^{-1} = f(x)$

To solve put y = x in f (x)

$x = (x + 1)^2 - 1$

$x + 1 = (x + 1)^2$

$x = -1, x = 0$

$x = \{0 -1\}$ Statement (1) is also correct

Q3. Let $f: \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \rightarrow R$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then

(A) f (x) is an odd function (B) f (x) is a one-one function

(C) f (x) is an onto function (D) f (x) is an even function

Ans. A, B, C

Sol. $f(x) = \frac{1}{\sqrt{x| - x}}$

For domain of real function

$|x| - x > 0$

$|x| > x$

$x \in (-\infty, 0)$

Q4. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta - \alpha)$ is______

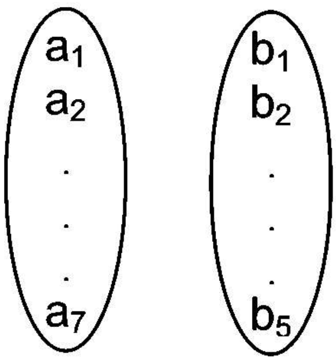
Ans. 119

Sol. n(X) = 5

n(Y) = 4

$\alpha \to$ Number of one-one function = ${}^{7}C_5 \times 5! = 21 \times 20 = 2520$

$\beta \to$ Number of onto function Y to X



1, 1, 1, 3 1, 1, 1, 2, 2

$$\frac{7!}{3!\,4!} \times 5! + \frac{7!}{(2!)^3\,3!} \times 5! = ({}^{7}C_3 + 3.\,{}^{7}C_3)\,5! = 4 \times {}^{7}C_3 \times 5!$$

$$\frac{\beta - \alpha}{5!} = 4 \times {}^{7}C_3 - {}^{7}C_5 = 4 \times 35 - 21 = 119$$

Q5. If the function f : R → R is defined by f(x) = |x|(x – sin x), then which of the following statements is TRUE?

(A) f is one-one, but NOT onto (B) f is onto, but NOT one-one

(C) f is BOTH one-one and onto (D) f is NEITHER one-one NOR onto

Ans. C

Sol. $\begin{cases} x(x-\sin x) & x \geq 0 \\ -x(x-\sin x) & x < 0 \end{cases}$

$\forall\, x \geq 0$

$(x) = 2x - (\sin x + x\cos x)$

$= (x - \sin x) + x(1 - \cos x)$ ³ 0

$f(x)$ is increasing

$f(x))_{min} = f(0) = 0$

$x \to \infty \Rightarrow f(x) \to$

$\Rightarrow \forall\, x \geq 0 \Rightarrow f(x) \in [0, \infty)$

$\forall\, x < 0$

$\because$ f is odd Þ f (x) is also increasing rangs of (x) for $x < 0$

$f(x) \in (-\infty, 0)$

Rangs of $f(x) = (-\infty, \infty) \Rightarrow f(x)$ is into

$f(x)$ is increasing $\forall x \in R \Rightarrow f$ is one – one

Q6. Let the function $f : [0, 1] \to R$ be defined by $f(x) = \frac{4^x}{4^x + 2}$. Then the value of $f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + \ldots\ldots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$, is given by λ then $(\lambda - 10)$ will be equal to

Ans. 9

Sol. $f(x) = \frac{4^x}{4^x + 2}$; Replace $x \to 1 - x$

$$f(1-x) = \frac{4^{1-x}}{4^{1-x} + 2} = \frac{\frac{4}{4^x}}{\frac{4}{4^x} + 2} = \frac{2}{4^x + 2}$$

$$f(x) + f(1-x) = \frac{4^x}{4^x + 2} = \frac{2}{4^x + 2} = 1 \forall x$$

So, $f\left(\frac{1}{40}\right) + f\left(\frac{39}{40}\right) = f\left(\frac{2}{40}\right) + f\left(\frac{38}{40}\right) = \ldots = f\left(\frac{19}{40}\right) + f\left(\frac{21}{40} = 1\right.$

$$\Rightarrow f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + \ldots f\left(\frac{19}{40}\right) + f\left(\frac{20}{40}\right) + f\left(\frac{21}{40}\right) + \ldots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) = 19 + f\left(\frac{20}{40}\right) - f($$

$= 19$

Here $\lambda = 19$ Hence $\lambda - 10 = 9$.

Q7. Let |M| denote the determinant of a square matrix. M. Let $g : \left[0, \frac{\pi}{2}\right] \to R$ be the function defined by $g(\theta) = \sqrt{f(\theta) - 1} + \sqrt{f\left(\frac{\pi}{2} - \theta\right) - 1}$ where

$$f(\theta) = \frac{1}{2}\begin{vmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix} + \begin{vmatrix} \sin\pi & \cos\left(\theta + \frac{\pi}{4}\right) & \tan\left(\theta - \frac{\pi}{4}\right) \\ \sin\left(\theta - \frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right) \\ \cot\left(\theta + \frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi \end{vmatrix}$$

. Let p(x) be a quadratic polynomial whose roots are the maximum and minimum values of the function $g(\theta)$, and $p(2) = 2 - \sqrt{2}$. Then, which of the following is/are TRUE ?

(A) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$ (B) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$ (C) $p\left(\frac{5\sqrt{2}-1}{4}\right)>0$

(D) $p\left(\frac{5-\sqrt{2}}{4}\right)<0$

Ans. A, C

Sol.
$$f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\frac{\pi}{4} & \tan\pi\end{vmatrix}$$

$$f(\theta)=\frac{1}{2}\begin{vmatrix}2 & \sin\theta & 1\\ 0 & 1 & \sin\theta\\ 0 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}0 & -\sin\left(\theta-\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & 0 & \log_e\left(\frac{4}{\pi}\right)\\ -\tan\left(\theta-\frac{\pi}{4}\right) & -\log_e\left(\frac{4}{\pi}\right) & 0\end{vmatrix}$$

$f(\theta) = (1 + \sin^2\theta) + 0$ (skew symmetric)

$$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$$

$= |\sin\theta| + |\cos\theta$ for $\theta \in \left[0, \frac{\pi}{2}\right]|$

$g(\theta)\in\left[1,\sqrt{2}\right]$

Again let $P(x) = k(x-\sqrt{2})(x-1)$

$2-\sqrt{2} = k(2-\sqrt{2})(2-1)$

$\Rightarrow k = 1$ ($P(2) = 2-\sqrt{2}$ given)

$\therefore P(x) = \left(x-\sqrt{2}\right)(x-1)$

for option P $\left(\frac{3+\sqrt{2}}{4}\right) < 0$ correct

option (B) P P $\left(\frac{1+3\sqrt{2}}{4}\right) < 0$ incorrect

option (C) P $\left(\frac{5\sqrt{2}-1}{4}\right) > 0$ correct

option (D) P $P\left(\frac{5-\sqrt{2}}{4}\right) > 0$ incorrect

Q8. Let f : R → R be a function defined by $f(x)=\begin{cases}x^2\sin\left(\frac{\pi}{x^2}\right), & \text{if} x\neq 0,\\ 0, & \text{if} x=0\end{cases}$. Then which of the following statements is TRUE?

(A) $f(x) = 0$ has infinitely many solutions in the interval $\left[\frac{1}{10^{10}},\infty\right)$

(B) $f(x) = 0$ has no solutions in the interval $\left[\frac{1}{\pi},\infty\right)$

(C) The set of solutions of $f(x) = 0$ in the interval $\left(0,\frac{1}{10^{10}}\right)$ is finite

(D) f(x) = 0 has more than 25 solutions in the interval $\left(\frac{1}{\pi^2}, \frac{1}{\pi}\right)$

Ans. D

Sol. for x > 0

$f(x) = 0 \Rightarrow x^2 \sin\left(\frac{\pi}{x^2}\right) = 0$

$\Rightarrow \sin\left(\frac{\pi}{x^2}\right) = 0$

$\Rightarrow \frac{\pi}{x^2} = n\pi$

$\Rightarrow x = \pm\frac{1}{\sqrt{n}}$

$x \in \left(\frac{1}{\pi^2}, \frac{1}{\pi}\right) \quad \Rightarrow \frac{1}{\pi^2} < \frac{1}{\sqrt{n}} < \frac{1}{\pi}$

$\pi < \sqrt{n} < \pi^2$

$\pi^2 < n < \pi^4$

f(x) = 0 has more than 25 solutions

Q9. Let f : R $\to$ R be a fuction such that f(x + y) = f(x) + f(y) for all x, y $\in$ R, and g : R $\to$ (0, ∞) be a function such that g(x + y) = g(x) g(y) for all x, y $\in$ R. If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ is ________.

Ans. 51

Sol. f : R $\to$ R

f(x + y) = f(x) + f(y) for all x, y $\in$ R

$\Rightarrow$ f(x) = ax

Now $f\left(\frac{-3}{5}\right) = 12$

a = – 20

$\therefore$ f(x) = – 20x

Again, g (x + y) = g(x). g(y) x, y $\in$ R.

$\Rightarrow$ f(x) = k^x

Now $g\left(\frac{-1}{3}\right) = 2$

$2 = k^{\frac{-1}{3}}$

$\Rightarrow \left(2^{-1}\right)^3 = k$

$k = \frac{1}{8}$

$\therefore \ g(x) = \left(\frac{1}{8}\right)^x$

Now, $\left(f\left(\frac{1}{4}\right)+g(-2)-8\right)g(0)$

$\Rightarrow [-5 + 64 - 8] . 1 \quad \Rightarrow 64 - 13 \quad \Rightarrow 51$

Q10. **PARAGRAPH :**

Let S = {1, 2, 3, 4, 5, 6} and X be the set of all relations R from S to S that satisfy both the following properties:

i. R has exactly 6 elements.

ii. For each (a, b) $\in$ R, we have $|a - b| \geq 2$.

Let Y = {R $\in$ X : The range of R has exactly one element } and

Z = {R $\in$ X : R is a function from S to S}.

Let n(A) denote the number of elements in a set A.

(There are two questions based on PARAGRAPH "I")

If $n(X) = {}^{m}C_6$, then the value of m is ________.

Ans. 20

Sol. All possible ordered pair (a, b) s.t. $|a - b| \geq 2$

(1, 3)	(2, 4)	(3, 1)	(4, 1)	(5, 1)	(6, 1)
(1, 4)	(2, 5)	(3, 5)	(4, 2)	(5, 2)	(6, 2)
(1, 5)	(2, 6)	(3, 6)	(4, 6)	(5, 3)	(6, 3)
(1, 6)					(6, 4)

$n(X) = {}^{20}C_6 \Rightarrow m = 20$

Q11. Let S = {1, 2, 3, 4, 5, 6} and X be the set of all relations R from S to S that satisfy both the following properties:

i. R has exactly 6 elements.

ii. For each (a, b) $\in$ R, we have $|a - b| \geq 2$.

Let Y = {R $\in$ X : The range of R has exactly one element } and

Z = {R $\in$ X : R is a function from S to S}.

Let n(A) denote the number of elements in a set A.

(There are two questions based on PARAGRAPH "I")

If the value of n(Y) + n(Z) is k^2, then |k| is ________.

Ans. 36

Sol. All possible ordered pair (a, b) s.t. $|a - b|$ ³ 2

(1, 3)	(2, 4)	(3, 1)	(4, 1)	(5, 1)	(6, 1)
(1, 4)	(2, 5)	(3, 5)	(4, 2)	(5, 2)	(6, 2)
(1, 5)	(2, 6)	(3, 6)	(4, 6)	(5, 3)	(6, 3)
(1, 6)					(6, 4)

n(Y) = 0 as maximum 4 ordered pairs are possible having same image

$n(Z) = 4 \times 3 \times 3 \times 3 \times 3 \times 4 = 1296$

$n(Y) + n(Z) = 1296 = (36)^2$

$\Rightarrow |k| = 36$

Exercise C1

Q1. The mapping $f : R \rightarrow R$ given by $f(x) = x^3 + ax^2 + bx + c$ is a bijection if

(A) $b^2 \leq 3a$ (B) $a^2 \leq 3b$ (C) $a^2 \geq 3b$ (D) $b^2 \geq 3a$

Ans. B

Sol. $f : R \rightarrow\rightarrow \rightarrow R$, $f(x) = x^3 + ax^2 + bx + c$

$f'(x) = 3x^2 + 2ax + b$

$D \leq 0$ or $4a2 - 12 \leq 0\frac{1}{2}$

or $a^2 \leq 3b$

Q2. If $f : [0, \infty) \rightarrow [0, \infty)$, and $f(x) = \frac{x}{1+x}$, then f is:

(A) one-one and onto (B) one-one but not onto (C) onto but not one-one

(D) neither one-one nor onto

Ans. B

Sol. $f : [\,0, \infty) \rightarrow [\,0, \infty)$ $\Rightarrow$ $f(x) = \frac{x}{x\ +\ x}$

$\frac{x_1}{1\ +\ x_1} = \frac{x_2}{1\ +\ x_2}$ $\Rightarrow$ $x_1 = x_2$ only

for given domain

$f(x) < 1$ $\therefore$ function is into

Q3. Suppose $f(x) = (x + 1)^2$ for $x \geq -1$. If g (x) is the function whose graph is the reflection of the graph of f (x) with respect to the line y = x, then g (x) equals:

(A) $-\sqrt{x} - 1, x \geq 0$ (B) $\frac{1}{(x+1)^2}, x > -1$ (C) $\sqrt{x+1}, x > -1$ (D) $\sqrt{x} - 1, x \geq 0$

Ans. D

Sol. -

Q4. Let function $f : R \rightarrow R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is:

(A) one to one and onto (B) one to one but not onto (C) onto but not one to one

(D) neither one to one nor onto

Ans. A

Sol. -

Q5. Let g (x) = 1 + x - [x] & f (x) = $\begin{cases} -1 & , x < 0 \\ d & , x = 0 \\ 1 & , x > 0 \end{cases}$. Then for all x, f (g (x)) is equal to

(A) x (B) 1 (C) f(x) (D) g(x)

Ans. B

Sol. n (into + onto) = 2^4

n(into) = 2

n (onto) = 16 – 2 = 14

Q6. If f: [1, ∞) → [2, ∞) is given by, f (x) = x + $\frac{1}{x}$, then f^{-1} (x) equals:

(A) $\frac{x+\sqrt{x^2-4}}{2}$ (B) $\frac{x}{1+x^2}$ (C) $\frac{x-\sqrt{x^2-4}}{2}$ (D) $1-\sqrt{x^2-4}$

Ans. A

Sol. f : [1, ∞) → [2, ∞)

y = f (x) = x + $\frac{1}{x}$ ⇒ x^2 – xy + 1 = 0 ⇒ $\frac{y \pm \sqrt{y^2-4}}{2}$

f^{-1} (x) = $\frac{y + \sqrt{x^2-4}}{2}$ as f^{-1} : [2, ∞)

Q7. Let f (x) = $\frac{\alpha x}{x+1}$, x ≠ –1. Then for what value of α is f (f (x)) = x?

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 1 (D) – 1

Ans. D

Sol. f(x) = $\frac{\alpha x}{x+1}, x \neq -1$

Now f(f(x)) x ⇒ f(x) = f^{-1} (x)

Let y = $\frac{\alpha x}{x+1}$ ⇒ xy + y = αx

⇒ x(y – α) = – y ⇒ x = $\frac{-y}{y-\alpha}$

f^{-1} (x) - $\frac{-x}{x-\alpha}$

Now = $\frac{\alpha x}{x+1} = \frac{-x}{x-\alpha}$

on solving we get α = – 1

Q8. If the function f: [1, ∞) → [1, ∞) is defined by f(x) = $2^{x(x-1)}$, then f^{-1}(x) is:

(A) $\left(\frac{1}{2}\right)^{x(x-1)}$ (B) $\frac{1}{2}(1+\sqrt{1+4\log_2 x})$ (C) $\frac{1}{2}(1-\sqrt{1+4\log_2 x})$ (D) not defined

Ans. B

Sol. $f(x) = 2^{x(x-1)}$

It is one-one onto function

$\log_2 y = (x-1)$

$\Rightarrow\ x^2 - x - \log_2 y = 0$

$x = \dfrac{1 \pm \sqrt{1 + 4\log_2 y}}{2}$

$f^{-1}(x) = \dfrac{1 + \sqrt{1 + 4\log_2 y}}{2}$

Q9. Let $f(x) = (x+1)^2 - 1,\ (x \geq -1)$. Then the set $S = \{x : f(x) = f^{-1}(x)\}$ is, if f is onto:

(A) $\left\{0, -1\dfrac{-3+i\sqrt{3}}{2}, \dfrac{-3-i\sqrt{3}}{2}\right\}$ (B) $\{0, 1, -1\}$ (C) $\{0, -1\}$ (D) empty

Ans. C

Sol. -

Q10. Let $f(x) = \sin x$ and $g(x) = |\ln x|$ if composite functions fog(x) and gof (x) are defined and have ranges R_1 & R_2 respectively then.

(A) $R_1 = \{u: -1 < u < 1\}$ $R_2 = \{v: 0 < v < \infty\}$

(B) $R_1 = \{u: -\infty < u \leq 0\}$ $R_2 = \{v: -1 < v < 1\}$

(C) $R_1 = \{u: 0 \leq u < \infty\}$ $R_2 = \{v: -1 < v < 1;\ v \neq 0\}$

(D) $R_1 = \{u: -1 \leq u \leq 1\}$ $R_2 = \{v: 0 < v < \infty\}$

Ans. D

Sol. -

Q11. If the functions f (x) and g (x) are defined on $R \rightarrow R$ such that

$f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}$, $g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}$

then $(f - g)(x)$ is

(A) one-one and onto (B) neither one-one nor onto (C) one-one but not onto

(D) onto but not one-one

Ans. A

Sol. $\phi(x) = f(x) - g(x)$

$= \begin{cases} -x & x \in Q \\ x & x \notin Q \end{cases}$

Q12. Which of the following statements are incorrect?

I If $f(x)$ and $g(x)$ are one to one then $f(x) + g(x)$ is also one to one.

II If $f(x)$ and $g(x)$ are one-one then $f(x) \cdot g(x)$ is also one-one.

III If $f(x)$ is odd then it is necessarily one to one.

(A) I and II only (B) II and III only (C) III and I only (D) I, II and III

Ans. D

Sol. -

Q13. Which of the following equations have the same graphs?

I. $y = x - 2$ II. $y = \dfrac{(x^2 - 4)}{(x + 2)}$ III. $(x + 2)y = x^2 - 4$

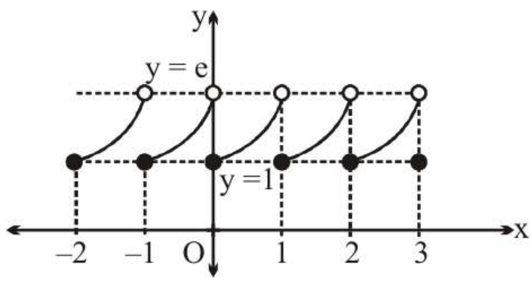
(A) I and II only. (B) I and III only. (C) II and III only.

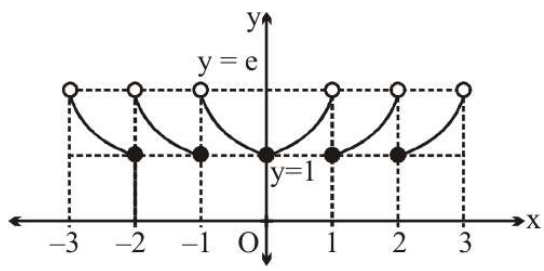
(D) All the equations have different graphs.

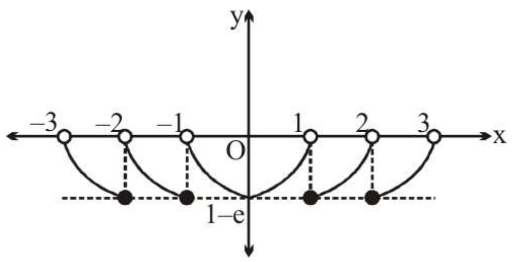
Ans. D

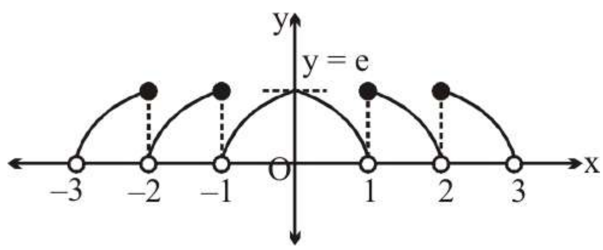
Sol. -

Q14. Which one of the following best represent the graph of function $f(x) = e^{\{|x|\}}$.
[Note: $\{\alpha\}$ denotes the fractional part of α.]

(A)


(B)


(C)


(D)


Ans. B

Sol. -

Q15. Set A consists of 4 distinct and set-B consists of 3 distinct elements. Number of functions that can be defined from a set A → B which are not onto is

(A) 35 (B) 45 (C) 63 (D) 9

Ans. B

Sol. -

Q16. Let f : R → R be defined as f(x) = $\sqrt[3]{[x]}$ and g : R → R is defined by g(x) = $\dfrac{1-x^2}{1+x^2}$, then which of the following statement(s) is (are) correct ?

(A) f is surjective (B) f(x) + g(x) is neither even nor odd (C) Range of g(x) is [–1, 1]

(D) f is bounded and g is unbounded

Ans. A, B, C, D

Sol. -

Q17. Let f(x) = $[x]^2 + [x+1] - 3$, where [x] denotes greatest integer less than or equal to x, then which of the following statement(s) is/are Correct?

(A) f(x) is many one function. (B) f(x) vanishes for atleast three values of x.

(C) f(x) is neither even nor odd function. (D) f(x) is aperiodic.

Ans. A, B, C, D

Sol. we have $f(x) = [x^2] + [x] - 2 \Rightarrow f(x) = ([x]+2)([x]-1) = \begin{cases} -2, & x \le x < 1 \\ 0, & 1 \le x < 2 \end{cases}$ and so on

$\Rightarrow f(x) = 0 \Rightarrow x \in [-2, -1) \cup [1, 2) \Rightarrow$ f(x) = 0 has infinite solutions.

Also $f(-x) \ne = f(x)$ and $f(-x) \ne -f(x) \Rightarrow$ f(x) is neither even nor odd function.

Now $f(x) = [x]^2 + [x] - 2$, where $[x]^3 + [x]$ is aperoidic.

$\Rightarrow$ f(x) is aperiodic.

Q18. Let f : [– 1, 1] onto [3, 5] be a linear polynomial. Which of the following can be true?

(A) $f\left(\frac{-1}{2}\right) = \frac{7}{2}$ (B) $f^{-1}\left(\frac{15}{4}\right) = \frac{1}{4}$ (C) $f(0) \ne 4$ (D) $f\left(\frac{1}{2}\right) + f\left(\frac{-1}{2}\right) = 8$

Ans. A, B, D

Sol. -

Q19. Let f : R → R and g : R → R be mappings such that f (g(x)) is an injective mapping, then which of the following statement(s) is(are) correct?

(A) g(x) must be an injective mapping.

(B) If g(x) is surjective then f(x) may not be injective mapping.

(C) If g(x) is not surjective, then f(x) may be injective or many-one mapping.

(D) f(x) may not be an injective mapping.

Ans. D, C, A

Sol. -

Q20. Let $f(x) = \begin{cases} x^2 - 4, & \text{if } |x| \le 3 \\ 5\,\text{sgn}|x-3|, & \text{if } |x| > 3 \end{cases}$

and $g(x) = 2\tan^{-1}(ex) - \frac{\pi}{2}$ for all $x \in R$, then which of the following is(are) correct?

[Note: sgn(k) denotes the signum function of k.]

(A) fog(x) is an even function. (B) gof (x) is an even function.

(C) gog (x) is an odd function (D) fof (x) is an odd function.

Ans. A, B, C

Sol. -

Q21. Let f(x) is a polynomial function defined R to R with leading coefficient 2 and degres 5 such that

$f(1) = 2$, $f(2) = 3$, $f(3) = 4$, $f(4) = 5$ and $f(5) = 6$, then which of the following hold(s) good ?

(A) $f(x)$ is many-one onto function (B) $f(x)$ is one-one onto function

(C) $f(6) = 127$ (D) $f(0) = -239$

Ans. A, D

Sol. -